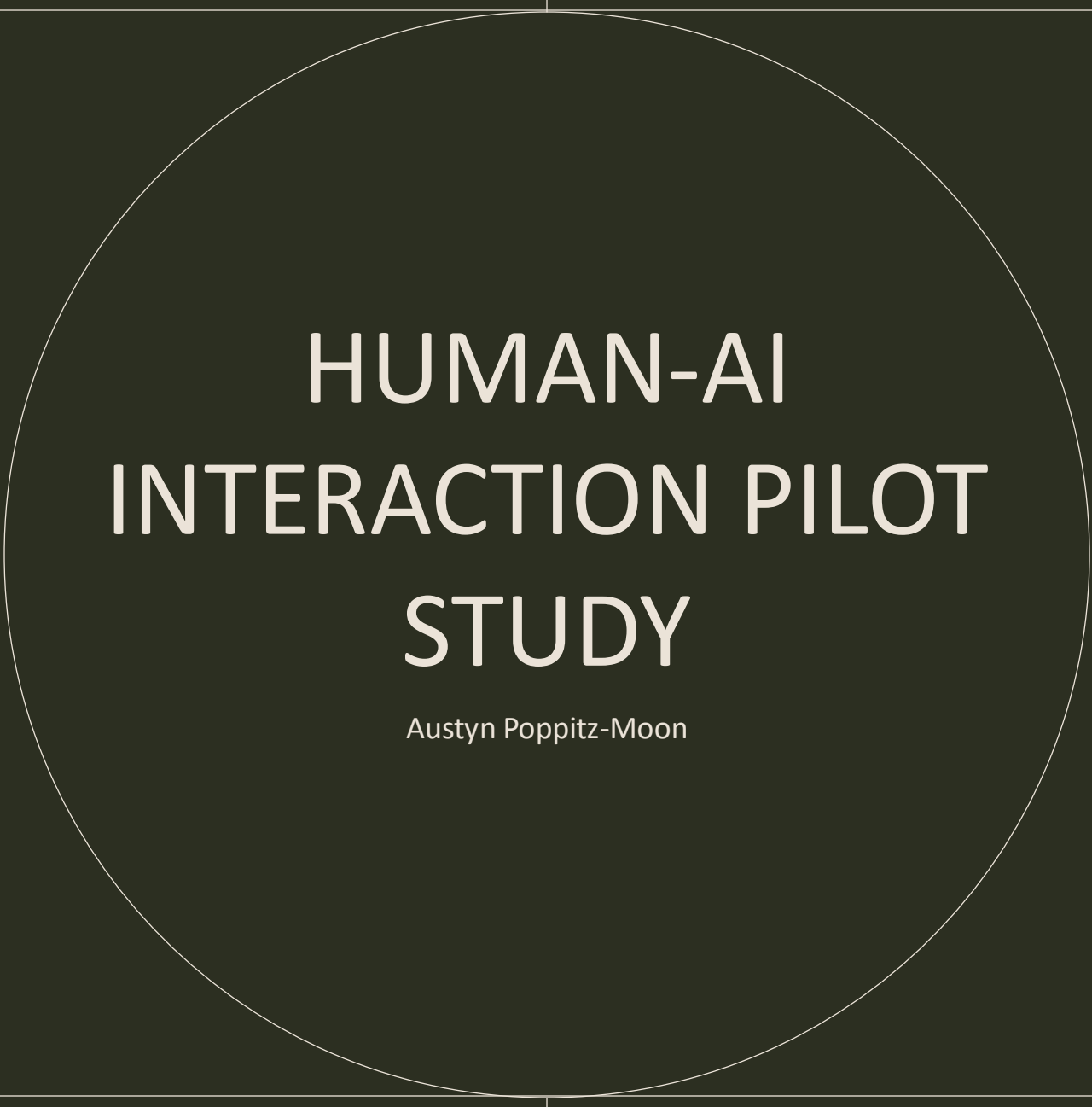




HUMAN-AI INTERACTION PILOT STUDY

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PILOT STUDY PURPOSE

- Research Questions:
 - How do students learn to AI tools?
 - There is a gap in research and understanding of the ways in which students currently interact with AI tools. Given that most schools are not explicitly teaching students how to interact with AI tools, they learn through unconventional sources.
 - How do students interact with AI tools?
 - Most instances of AI interaction within academia happens out of view of educators and academic spaces as AI use is generally frowned upon. Given that AI tools are shaping academic outcomes, it is important to understand to what extent the agency of an AI tool is having over a students work
 - How successful are AI tools at the tasks given to them by students?
 - Simply put, how helpful is AI for a student? Do students recognize when an AI tool is hallucinating? Or possibly taking them down the wrong path?

AI USE ASSESSMENT GUIDELINES

- **Session 1: Information Seeking**
 - The purpose of this session is to observe how you're able to gather information from an AI, so feel free to ask these questions in whatever way works best for you.
 - Ask the AI to explain a concept that is complex or unfamiliar to you.
 - Ask follow-up questions that are genuine to the conversation.
 - Ask the same question in two different ways to see if you get a better answer.
- **Session 2: Problem Solving/Decision Support**
 - Ask the AI to generate ideas/solutions to a specific assignment you have. For example, ideas on research topic for a final assignment within a given topic of study. The purpose of this session is to observe how you're able to problem solve, negotiate ideas and collaborate with AI tools.
 - Ask the AI to generate options in the form of research topics, solutions to a problem.
 - Ask for step by step instruction to achieve your goals (Outline for a paper, pseudocode for a coding project, etc.)
 - Ask the AI for its opinion on the options it has provided based on the information you have provided it with as well
- **Session 3: Process Comparison**
 - Complete an assignment without AI assistance. Once you have completed said assignment have AI complete the same assignment from start to finish. The purpose of this assignment is to compare the results of your own process versus the process of having AI complete an assignment for you.
 - Choose a straight forward assignment, likely a research paper or a coding assignment (I am open to whatever though, it would be interesting to see what AI can and can't do). Feel free to use assignments you have already completed as long as you remember doing it.

METHODS

- **Semi-Structured Interviews**

- Location: All semi-structured interviews took place online via zoom.
- Data Collected:
 - Pre-study interviews gathered information pertaining to a participants understanding of AI tools, their position at the university (Undergrad/Engineering)(Graduate/Social Science).
 - Post-study interviews gathered information pertaining to their experience with the study prompts guiding AI interactions that they conducted and overall experience with diary studies data collection to help inform future grad thesis work.
- Purpose: The purpose of using this method was to help build rapport with participants and to get feedback on their experience with the pilot study design.

- **Diary Studies**

- Location: All diary entries collected were audio recordings created by test participants after having completed each AI interaction session.
- Collected: The data received detailed the overall interaction with the AI and interaction guidelines. Audio files collected were transcribed for coding analysis. Participants also provided screenshots of their AI interaction.
- Purpose: The purpose of this method was to test out the reliability of having participants recount their experiences without the pressure of being watched by the researcher. Completing assignments or projects may not always be a linear process, by allowing them to record their experiences

FORMS OF DATA

- **Forms of Data:**
 - Diary Audio Files -> Converted into transcripts using AI features on Riverside software
 - AI Interaction Screenshots -> Converted to PDF for easy file transcription
 - Semi-structured Interviews -> Converted into transcripts using AI features on Riverside software
 - All transcripts were uploaded into qualitative data analysis software known as MaxQDA
 - With this software I was able to create codes to classify Human-AI interactions.

ANALYSIS: AI HALLUCINATIONS

- **Participant 2, AI Model: Ollama Gemma 3B Model**
 - **8. Links and other resources to answer prompt**
 - This code refers to moments where an AI tool provides the user with a link to an internet resource with the purpose of solving a problem or answering a question given to it by the user.

YouTube Tutorials - GamaKay TK75 Switch
Replacement: <https://m.youtube.com/watch?v=mJq-UvHIK-E>
GamaKay Support Website: <https://gamakay.com/> (Check for firmware updates or potential support)

Resources:

- Microsoft Support Article on Remapping the Windows Key: <https://support.microsoft.com/en-us/windows/remap-windows-key-2327bb58-a2f9-4d9a-8d90-32c4c6ef8695>

I sincerely apologize for the initial misinformation. It's critical to verify information, and I appreciate you pointing out the change in location. Let's start with trying the "Remap Keys" setting within the Keyboard Layout Settings. If that doesn't work, we'll move on to investigating the driver and software conflicts. Which of these steps would you like to try first?

- **46. False link hallucination**
 - This code refers to moments where the user identifies that a link that the AI tool has provided does not exist and was not prompted to give a link in its response. It did this on its own.
- **Excerpt From Diary Transcript:**
 - "which I told it was not there and it apologized a bunch of times which also didn't help and sent me somewhere else where it also didn't exist."
 - "It's worth noting that this model does not have access to the internet, and so it did not do any searches. This was just it kind of spinning its wheels."
- **Analysis:**
 - What I found most interesting about this code to include it in this analysis is that out of the 6 separate interactions with both of my participants, only the Gemma 3B hallucinated links that do not exist. When it was caught doing this it began to apologize in a way that felt incredibly urgent, as though it mattered. This type of apologizing and sycophantic type behavior also falls under the code "I am totally human", which describes instances where AI is mimicking human conversational conventions that would normally function to keep a conversation going. Based on the context of this interaction I would say that this model is highly tuned for user engagement, in that it prioritizes engagement over accuracy. It is striking that it would create links that do not exist, especially as it does not have the capability of even accessing the internet.

ANALYSIS: CONVERSATIONAL DIRECTION-ING

- **21. Conversational Direction-ing**

- This code refers to moments where the AI presents users with possible follow-up questions and process directions related to prompts provided by the user.

- **Open AI's ChatGPT Participant 1**

- "If you'd like, I can next: derive the turbine efficiency equation step-by-step, connect turbines to T-s diagrams (super helpful visually), or explain blade shapes and why impulse vs reaction stages matter. Just tell me which direction you want to go next."

- **Open AI's ChatGPT Participant 1**

If you'd like, I can also:

- Convert these into **measurable engineering requirements**
- Adjust them for **beginner vs. professional users**
- Help with a **needs hierarchy** or **House of Quality**

- **Ollama's Gemma 3B Participant 2**

Do you find this explanation reasonable, or would you like me to elaborate on any specific point?

- Conversational direction-ing appeared to be the most prevalent amongst the interactions between participant 1 and ChatGPT. While participant 2 only had a single instance of this type of behavior. The main difference between the prompts provided by each participant was that participant 1's prompts were very brief and lacked detail in terms of describing what type of output was expected, so the AI seemed to take more liberties in determining what direction the conversation was going to take. Participant 2's prompts followed a pattern where the expected output was clearly defined, and while the AI that participant 2 used did not always produce the correct answer, it did not steer the conversation as often. I am not sure if this is model dependent or user dependent but still points to ways in which user interactions can shape the level of agency that an AI has over the way users interact with information that they may or may not understand.